

Roll No.

TCS-307

B. TECH. (CSE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022

OBJECT ORIENTED PROGRAMMING WITH C++

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Illustrate the features of object oriented programming paradigm with the help of examples. (CO1)

OR

- (b) With the help of a program, demonstrate the need of namespace in C++. (CO1)

2. (a) Implement a C++ program that takes an alphanumeric string as an input, remove all the characters of the string excluding the numeric : (CO1)

Input string : abc34d5bb7uu2u7q8ww2

Typical Output : 34572782

OR

- (b) Implement a C++ program to find the most occurring element in an array of integers : (CO1)

Input : 5 7 9 2 5 1 5 2 5

Typical output : 5

P. T. O.

3. (a) Discuss the need of constructor in C++. With the help of a C++ code, explain different types of constructors in C++. (CO2)

OR

- (b) Implement a C++ program to swap the value of data members present in two different classes with the help of friend function. (CO2)
4. (a) Implement a class Book with the following specifications : (CO2)

Private members :

Book_no : integer type

Book_Title : 20 characters

Price : float (price per copy)

totalCost() : A function to calculate the total cost for N number of copies where N is passed to the function as argument.

Public members :

input() : function to read Book_no, Book_title and Price

purchase() : function to ask the user to input the number of copies to be purchased.

It invokes totalCost() and prints the total cost to be paid by the user.

OR

- (b) Implement a class Batsman with the following specifications and create an object that calls the members and displays the necessary output : (CO2)

Private members :

bcode : 4 digits code (number)

bname : 20 characters

(3)

innings, notout, runs (integer type)

batavg : double type

batavg : it is calculated according to the formula :

$$\text{batavg} = \text{runs} / (\text{innings} - \text{notout})$$

calcAvg() : function to compute batavg

Public members :

readData() : Function to accept value from bcode, name, innings,
notout

showData() : Function to display the data members on the screen.

5. (a) Implement a C++ program to overload post decrement operator (— —) with the help of friend function. (CO3)

OR

- (b) Discuss the need of operator overloading in C++. Write a program in C++ to overload the binary operator of the type $\text{ob2} = 10 + \text{ob1}$, where ob1 is an object of class Complex that contains two arguments (real, imaginary), here 10 is interpreted as a real part only ($10+0*i$) i.e., no imaginary part. The sequence of the statement $\text{ob2} = 10 + \text{ob1}$ should not be changed. (CO3)